

Capture the Flag 7: Stop Putting Everything on One Screen

Goal: Split your Student Directory into multiple screens using Flutter navigation, while using a modal for editing student information.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

Screenshot:

- Terminal showing your current branch

```
C:\Users\johnf\Downloads\ggs>git switch -c feature/multi-screen-navigation
Switched to a new branch 'feature/multi-screen-navigation'
```

```
C:\Users\johnf\Downloads\ggs>git branch
feature/conditional-ui
feature/data-ui
* feature/multi-screen-navigation
feature/student-interactions
feature/student-list
feature/widgets
main
```

🚩 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Create a dedicated **Student List Screen** containing your existing student list.

Research how Flutter creates separate screens using **Widget** classes.

Test

The Student Directory should function as its own screen.

Screenshot:

- Student List Screen
- Student List screen code
- Application showing the screen

```

123 // STUDENT LIST SCREEN
124 class StudentListScreen extends StatefulWidget {
125   const StudentListScreen({super.key});
126
127   @override
128   State<StudentListScreen> createState() =>
129     _StudentListScreenState();
130 }
131
132 class _StudentListScreenState extends State<StudentListScreen> {
133   // FLAG 4 & FLAG 5 - LOADING STATE
134   bool isLoading = true;

```

Student Directory

Ivan Villareal

ACTIVE

BSIT - 3rd Year

Age: 24

Hobby: Photography, Videography, & Video Editing

Student ID: 2026-0001

Email: ivan.villareal@dbtc-cebu.edu.ph

Favorite Subject: Snacks

Favorite
 Edit
 Delete

Mannes Artajo

INACTIVE

⚠ This student is currently inactive.

BSIT - 2nd Year

Age: 21

Hobby: Reading

Student ID: 2026-0002

Email: mannes.artajo@dbtc-cebu.edu.ph

Favorite Subject: Programming

🚩 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

When a student is selected, navigate:

Student List → Student Details

Research Flutter's **Navigator** and how to move between screens.

Test

1. Select a student.
2. Open the Student Details Screen. (No data displayed yet)
3. Return to the Student List using back navigation.

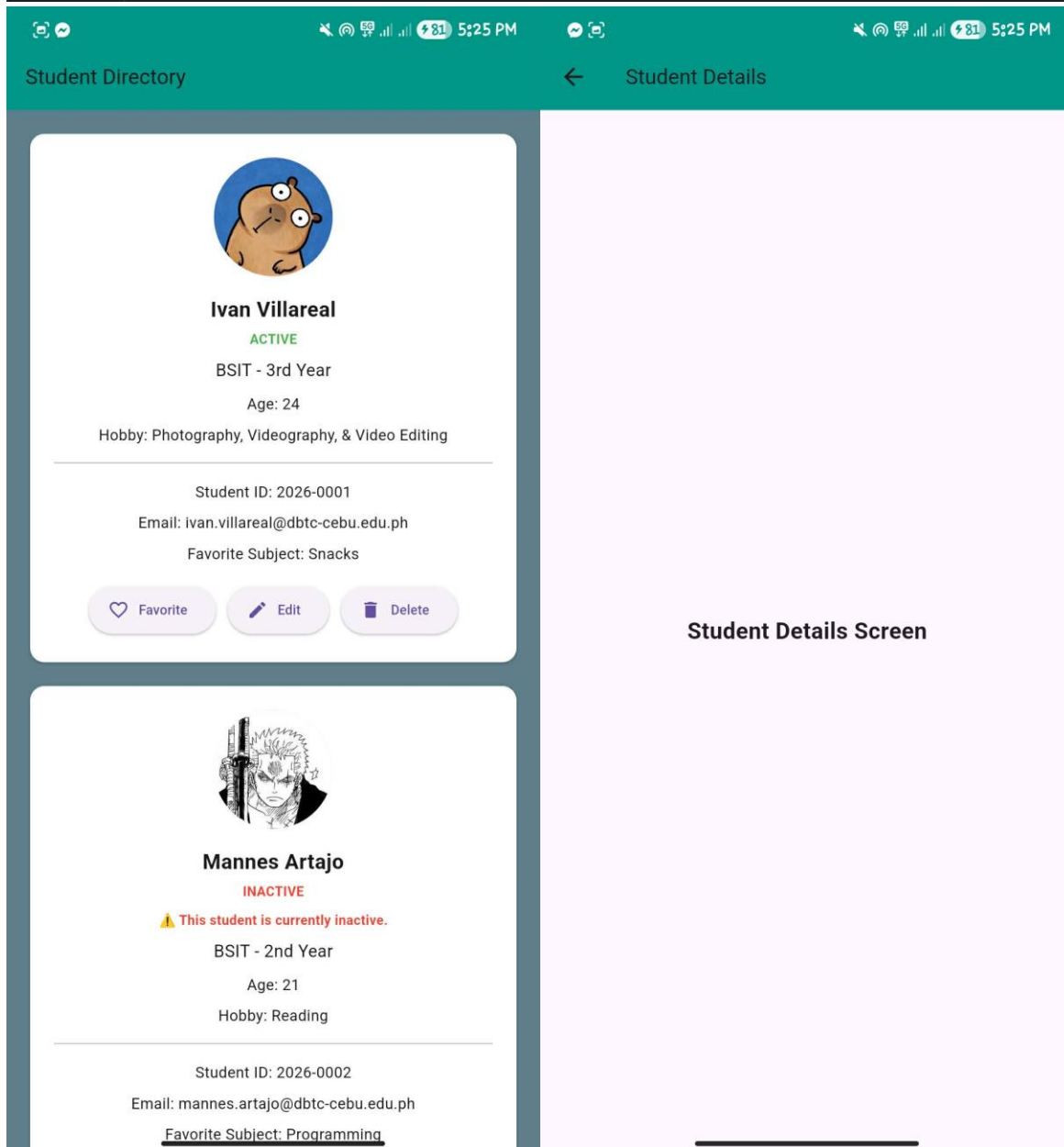
Screenshot:

- Student List Screen
- Student Details Screen
- Navigation code

```

341 //CAPTURE THE FLAG 7 #2 NAVIGATE TO STUDENT DETAILS
342 void _openStudentDetails() {
343     Navigator.push(
344         context,
345         MaterialPageRoute(
346             builder: (context) => const StudentDetailsScreen(),
347             MaterialPageRoute
348         );
349     }

```



🚩 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

The Details Screen should display the correct information for the selected student.

Display relevant information such as:

- Name
- Student ID
- Course
- Status
- Other existing student data

Research how data can be passed between Flutter screens.

Test

Selecting different students should display their corresponding information.

Student A → Student A's Details

Student B → Student B's Details

Screenshot:

- Selected student
- Student Details
- Data-passing code

```

341 //CAPTURE THE FLAG 7 #FLAG 3 - NAVIGATE AND PASS STUDENT DATA
342 void _openStudentDetails(Student student) {
343     Navigator.push(
344         context,
345         MaterialPageRoute(
346             builder: (context) => StudentDetailsScreen(
347                 student: student,
348             ), StudentDetailsScreen
349         ), MaterialPageRoute
350     );
351 }

```



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🚩 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Add an **Edit** action to the Student Details Screen.

When selected, display an editing interface using a **modal**, such as a dialog or bottom sheet.

Research Flutter's:

- `showDialog`
- `showModalBottomSheet`

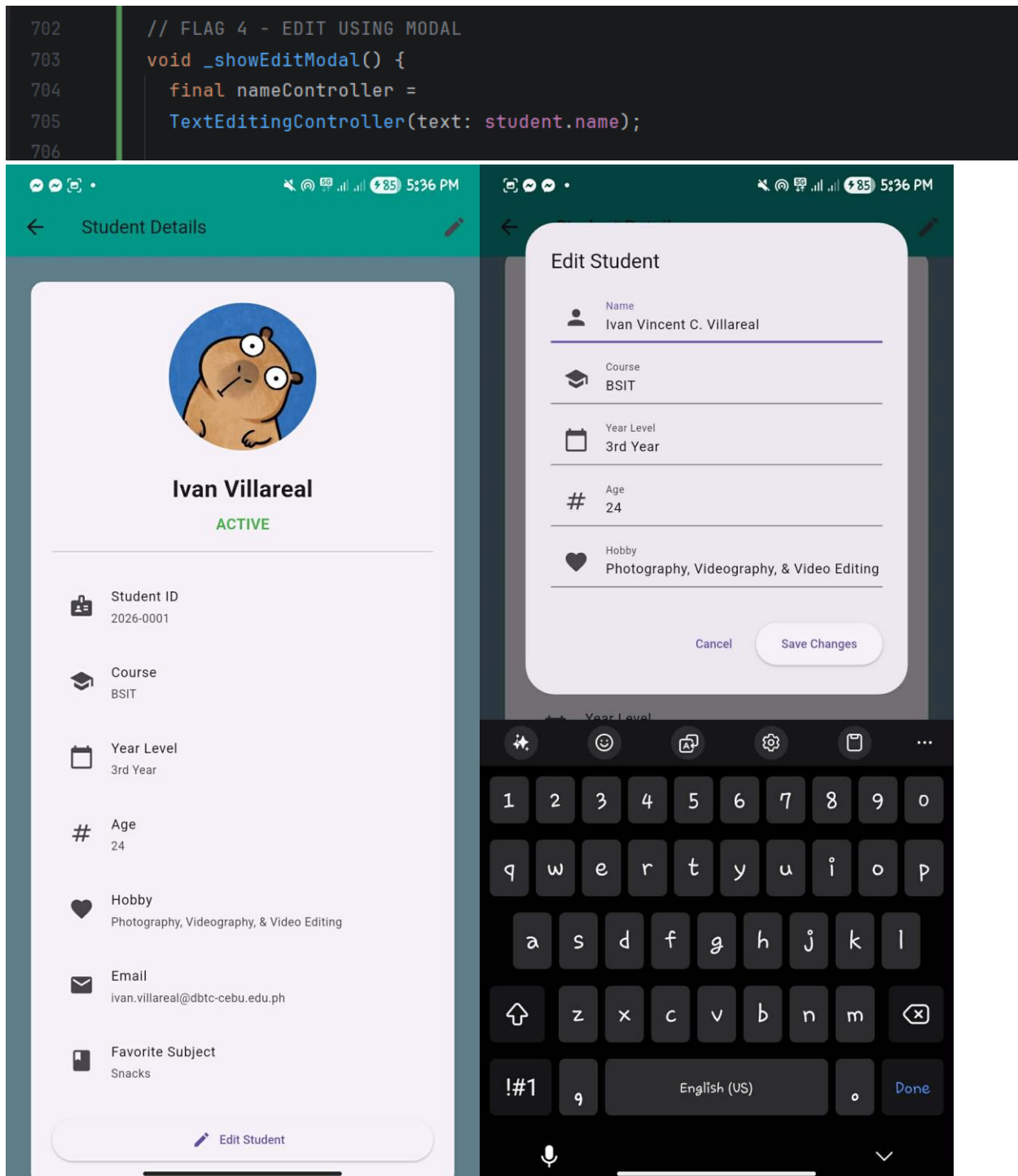
You may choose whichever approach is appropriate for your design.

Test

1. Open Student Details.
2. Select Edit.
3. An editing modal should appear.
4. The modal should display the appropriate student's information.
5. Close or submit the modal.

Screenshot:

- Student Details Screen
- Edit modal
- Modal code



🚩 Flag 5 — Create the Add Student Screen

Objective: Create a separate screen for adding a new student.

Add an action from the Student List that navigates to:

Student List → Add Student

Create the **Add Student Screen**.

For now, focus on creating the screen and navigation structure.

Test

Selecting Add Student should open the Add Student Screen.

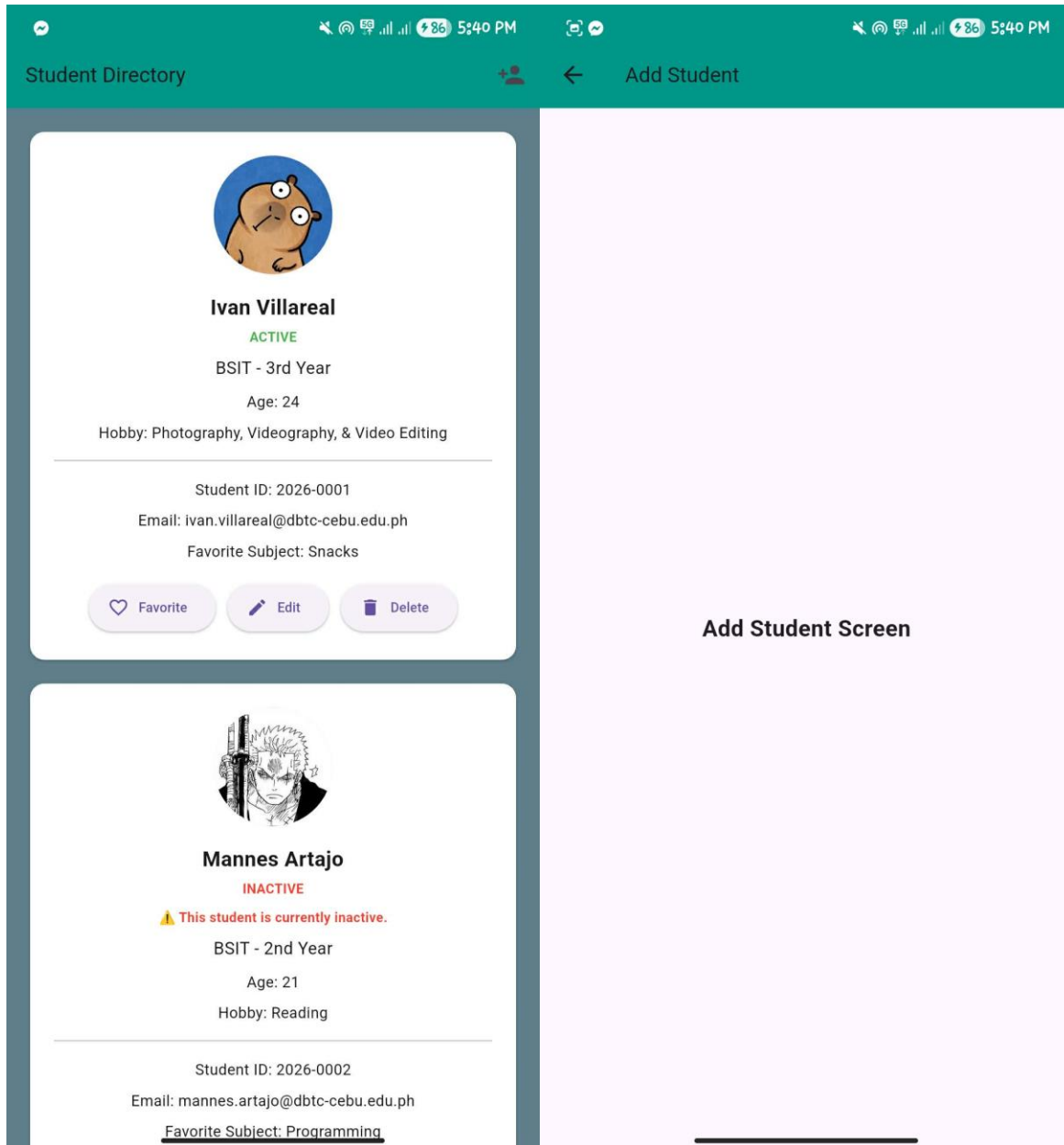
Screenshot:

- Student List with Add action
- Add Student Screen
- Navigation code

```

397 void _openStudentDetails(Student student) async {
398     final updatedStudent = await Navigator.push<Student>(
399         context,
400         MaterialPageRoute(
401             builder: (context) => StudentDetailsScreen(
402                 student: student,
403             ), StudentDetailsScreen
404         ), MaterialPageRoute
405     );

```



🚩 Flag 6 — Prepare the Add Student Screen

Objective: Prepare the Add Student Screen for future functionality.

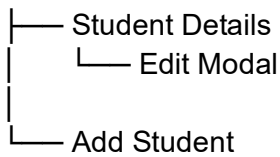
Organize the screen so it is ready to receive student input.

Do not implement the student input fields yet.

The screen should simply exist as a separate destination.

Your application should now have:

Student List



Test

You should be able to navigate between the appropriate screens without placing everything on one screen.

Screenshot:

- Student List
- Student Details
- Edit Modal
- Add Student Screen

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Student Directory

+ < Student Details



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BSIT • 3rd Year

ACTIVE

☆

✎

🗑



Mannes Artajo

BSIT • 2nd Year

INACTIVE

This student is currently inactive.

☆

✎

🗑



Wilken Montebon

BSIT • 1st Year

ACTIVE

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John Kevin Villacorte

BSIT • 4th Year

INACTIVE

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Keith Francheska Lopez

BSIT • 3rd Year

ACTIVE

☆

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Maria Santos

BSIT • 2nd Year

ACTIVE

☆

✎

🗑



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ACTIVE

Student ID: 2026-0001

Course: BSIT

Year Level: 3rd Year

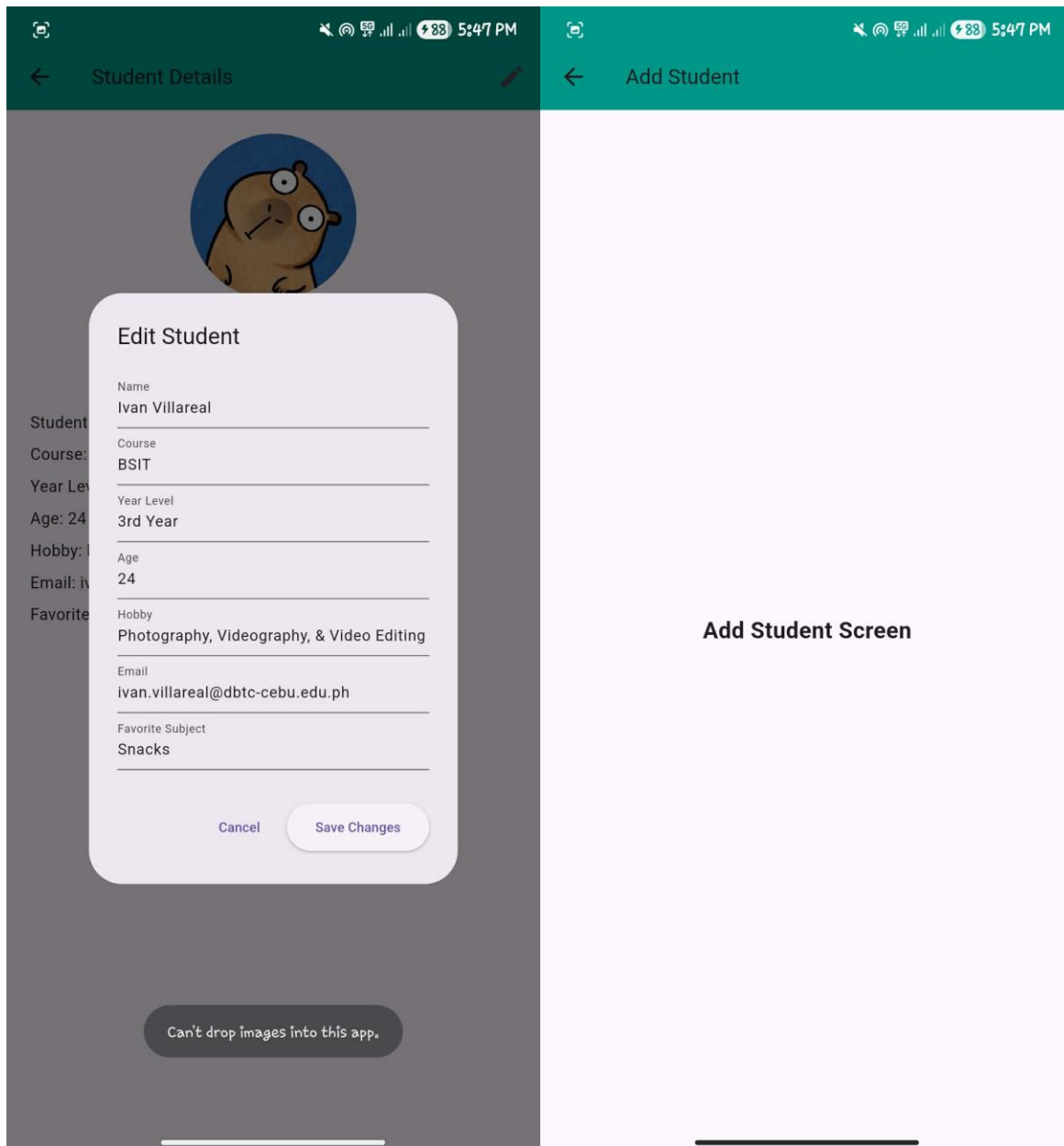
Age: 24

Hobby: Photography, Videography, & Video Editing

Email: ivan.villareal@dbtc-cebu.edu.ph

Favorite Subject: Snacks

✎ Edit Student



🚩 Flag 7 — Connect the Routes

Objective: Organize the application's screens using Flutter routes.

Create routes for the major screens:

/student-list
/student-details
/add-student

Research:

- Named routes
- `Navigator.pushNamed`
- Passing arguments through routes

The **Edit Modal** should not be a route.

Test

Verify that:

- Student List opens correctly.
- Student Details opens correctly.
- The correct student data is passed.
- Add Student opens correctly.
- Edit opens as a modal from Student Details.

Screenshot:

- Route configuration
- Navigation using routes
- Student Details receiving student data
- Edit modal

```

145 // NAMED ROUTES
146 initialRoute: "/student-list",
147
148 routes: {
149   "/student-list": (context) =>
150     const StudentListScreen(),
151     // FLAG 7
152     // RECEIVE STUDENT FROM ROUTE
153   "/student-details": (context) {
154     final student =
155       ModalRoute.of(context)!
156         .settings
157         .arguments as Student;
158
159
200 // OPEN STUDENT DETAILS
201 Future<void> _openStudentDetails(
202   Student student) async {
203   final updatedStudent =
204     await Navigator.pushNamed<Student>(
205       context,
206       "/student-details",
207       arguments: student,
208     );

```

```

152 // FLAG 7
153 // RECEIVE STUDENT FROM ROUTE
154 "/student-details": (context) {
155     final student =
156     ModalRoute.of(context)!
157     .settings
158     .arguments as Student;
159
160     return StudentDetailsScreen(
161     student: student,|
162     ); StudentDetailsScreen
163 },

```

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Student Directory

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 BSIT • 3rd Year
ACTIVE

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 BSIT • 2nd Year
INACTIVE
 This student is currently inactive.

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Maria Santos
 BSIT • 2nd Year
ACTIVE

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

feature/multi-screen-navigation

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
C:\Users\johnf\Downloads\ggs>git status
On branch feature/multi-screen-navigation
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   my_first_flutter_app/lib/main.dart

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\johnf\Downloads\ggs>git add .

C:\Users\johnf\Downloads\ggs>git status
On branch feature/multi-screen-navigation
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   my_first_flutter_app/lib/main.dart

C:\Users\johnf\Downloads\ggs>git commit -m "Feat: Complete CTF 7 multi-screen-navigation"
[feature/multi-screen-navigation ala5b91] Feat: Complete CTF 7 multi-screen-navigation
1 file changed, 780 insertions(+), 328 deletions(-)
```